

Inferential Models

A framework for valid and efficient imprecise-probabilistic inference

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Statisticians want numerical measures of the degree to which data support hypotheses. —Hacking

Perhaps the most important unresolved problem in statistical inference is the use of Bayes theorem in the absence of prior information. —Efron

As Bayes perceived, the concept of Mathematical Probability affords a means, in some cases, of expressing inferences... and of expressing them rigorously, in that the nature and extent of the uncertainty is specified with exactitude, yet it is by no means axiomatic that the appropriate inferences... should always be rigorously expressible in terms of this same concept. —Fisher

- Statistical inference is concerned with quantification of uncertainty about unknowns based on data, models, etc.
- The phrase “uncertainty quantification” is almost exclusively associated with (ordinary/precise) *probability*
- But this isn't the only option...
- In fact, there are clues in the literature that ordinary probability isn't right for statistical inference:
 - Fisher's *fiducial probability* lost out to Neyman's *confidence*
 - Bayesian inference has reliability guarantees only when the prior is *non-informative*, a notion foreign to probability
 - Dempster's efforts to “fix” Fisher's fiducial argument involved different ideas, i.e., *imprecise probability*

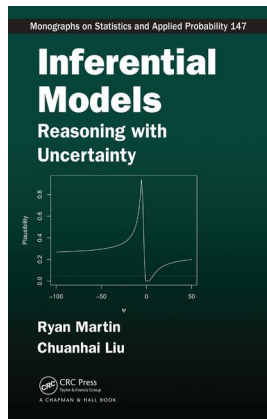
- So, despite statisticians' desire to solve the “most important unresolved problem” within the theory of probability, there's reason to believe this isn't possible
- What, then, is a more appropriate framework?
- Of course, it's not just a matter of switching our mode of UQ, the new formulation has to “work”
- Key questions:
 - for statistical inference, what does it mean for UQ to *work*?
 - how to construct something that *works* in the above sense?
- My/our goal here is to answer these questions

This series of lectures

- These lectures split (unequally) into three parts:
 - 1 *What and Why*
 - 2 *How*
 - 3 *What's next*
- More specifically:
 - 1 Why do I define “works” the way I do and what are the consequences of this definition?
 - 2 Given the goal to construct something that “works” in this sense, how to actually do the construction?
 - 3 Stuff we're currently thinking about

References

- Part 2 of the lecture is mostly from the book — you can find the corresponding papers on my website
- Parts 1 and 3 are based on some more recent papers



- statistical problem setup
- probabilistic inference
- validity property and its consequences
- questions:
 - can data-dependent probabilities be valid?
 - if not probabilities, then what?
- frequentism, validity, and imprecise probability
- concluding remarks

Statistical inference problem

- Observable data: Y in a sample space $\mathbb{Y}$
- Data are assumed to be relevant to the phenomenon under investigation, formalized through a *model*
- Statistical model: a collection of probability distributions for Y depending on a parameter $\theta \in \mathbb{T}$,

$$\mathcal{P} = \{P_{Y|\theta} : \theta \in \mathbb{T}\}$$

- Θ is the unknown or uncertain “true” model parameter, while θ and ϑ denote generic values thereof
- Goal is to quantify uncertainty about the unknown Θ based on observed value $Y = y$ of the data

- Two mainstream schools of thought:
 - *frequentist*
 - *Bayesian*
- These two approaches are very different:
 - the former constructs procedures with “good properties”
 - the latter uses a prior distribution for Θ and conditional probability theory to construct a posterior distribution
- Having two accepted theories is problematic on many levels, it suggests important connections/insights are missing
- My/our goal is to fill this gap

- Following Hacking, by “UQ” I mean a function that:
 - depends on data $Y = y$, assumed model, etc.
 - assigns to each hypothesis $A \subseteq \mathbb{T}$ a number in $[0, 1]$ that represents one's degree of support in A , given data, etc.
- A (super-additive) *capacity* γ is a set function that satisfies:
 - $\gamma(\emptyset) = 0$ and $\gamma(\mathbb{T}) = 1$
 - if $A \subseteq A'$, then $\gamma(A) \leq \gamma(A')$
 - $\gamma(A \cup A') \geq \gamma(A) + \gamma(A')$ for all $A \cap A' = \emptyset$

Definition.

An *inferential model* (IM) is a mapping from $(y, \mathcal{P}, \dots)$ to a data-dependent capacity $\underline{\Pi}_y : 2^{\mathbb{T}} \rightarrow [0, 1]$ such that, for each $A \subseteq \mathbb{T}$, $\underline{\Pi}_y(A)$ represents one's degree of support in A , given data, etc.

Definition — copied from above.

An *inferential model* (IM) is a mapping from $(y, \mathcal{P}, \dots)$ to a data-dependent capacity $\underline{\Pi}_y : 2^{\mathbb{T}} \rightarrow [0, 1]$ such that...

- Duality:³
 - $\underline{\Pi}_y$ has a dual, $\overline{\Pi}_y(A) = 1 - \underline{\Pi}_y(A^c)$, sub-additive
 - $\overline{\Pi}_y(A)$ represents the plausibility of A , given data, etc.
 - super-additivity implies $\underline{\Pi}_y(\cdot) \leq \overline{\Pi}_y(\cdot)$
- Examples include:
 - Bayesian posterior distributions
 - fiducial and confidence distributions
 - Dempster–Shafer
 - ...

³Probabilities are self-dual capacities

- The IM definition itself is too vague to be useful...
- To sharpen our focus, we need some additional constraints
- Basic question: *what do we want to use an IM for?*
- We don't really need an IM for:
 - point estimation
 - isolated hypothesis testing
 - ...
- The motivation is “uncertainty quantification,” getting an overall picture of where support is assigned
- That is, we care about the mapping $A \mapsto \underline{\Pi}_y(A)$

- One idea: require the IM to be *coherent*:⁴
 - subjective, internal consistency/rationality
 - can't be made a sure loser with prices $\underline{\Pi}_y(A)$
- But the statistics problem has non-subjective elements
- Assertions about Θ are *true* or *false*, and inferring a false assertion can have negative consequences
- Idea: require $\underline{\Pi}_y$ to prevent these bad outcomes...?

It is unacceptable if a procedure... of representing uncertain knowledge would, if used repeatedly, give systematically misleading conclusions. —Reid & Cox

⁴Requires that $A \mapsto \underline{\Pi}_y(A)$ is 2-monotone, minimal constraint

- $\underline{p}_y$ depends on data y that we can't control, so it's not possible to entirely avoid the bad outcomes
- However, we can prevent *systematically* bad outcomes
- Here's how we're going to use the IM:⁵

if $\underline{p}_y(A)$ is sufficiently large, then infer A

- The IM would be calibrated, i.e., inference isn't systematically misleading, if, roughly,

$$(\Theta \notin A \text{ and } Y \sim P_{Y|\Theta}) \implies \underline{p}_Y(A) \text{ tends to be small}$$

- We just need to make this more precise

⁵There's a set of precise probabilities lower-bounded by $\underline{p}_y$, so " $\underline{p}_y(A)$ is large" implies "all these probabilities are large" — more later

Definition.

An IM is *valid* if, for all $A \subseteq \mathbb{T}$ and all $\alpha \in [0, 1]$,

$$\sup_{\Theta \notin A} P_{Y|\Theta} \{ \underline{\Pi}_Y(A) > 1 - \alpha \} \leq \alpha$$

or, equivalently,

$$\sup_{\Theta \in A} P_{Y|\Theta} \{ \overline{\Pi}_Y(A) \leq \alpha \} \leq \alpha$$

- Intuitively, validity controls the chance that the IM
 - assigns relatively high support to false hypotheses
 - assigns relative low plausibility to true hypotheses
- Protects against “systematically misleading conclusions”

Theorem – statistical properties.

If the IM $(y, \mathcal{P}, \dots) \mapsto (\underline{\Pi}_y, \overline{\Pi}_y)$ is valid, then:

- 1 “reject $H_0 : \Theta \in A$ iff $\overline{\Pi}_y(A) \leq \alpha$ ” is a size α test
 - 2 $\{\theta : \overline{\Pi}_y(\{\theta\}) > \alpha\}$ is a $100(1 - \alpha)\%$ confidence region
- Validity implies that we immediately get statistical procedures with the desired frequentist properties
 - Makes sense: valid UQ ought to imply good procedures
 - More consequences of validity later...

- But how can validity be achieved?
- We need to understand what kind mathematical structures in $A \mapsto \overline{\Pi}_y(A)$ are compatible with validity
- In particular:
 - can ordinary/precise/additive probabilities achieve validity?
 - and, if not, then what can?
- Preview:
 - validity isn't compatible with probabilities
 - validity is compatible with a certain imprecise prob's

Satellite collision problem

- A satellite orbiting Earth could collide with another object
- Potential huge mess, so navigators try to avoid collision
- Data on position, velocity, etc is used, along with physical and statistical models, to compute a *collision probability*⁶
- If collision probability is low, then satellite is judged safe; otherwise, some action is taken
- An unusual phenomenon has been observed in the satellite conjunction analysis literature:

noisier data makes non-collision probability large

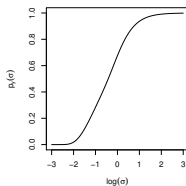
⁶Collision probability is based on a standard Bayes/fiducial/confidence distribution; see Balch, M., and Ferson, arXiv:1706.08565.

Satellite collision, cont.

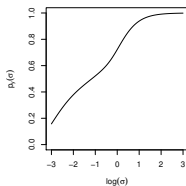
■ Simple illustration:

- $\|y\|$ denotes measured distance between satellite and object
- Suppose: true distance ≤ 1 implies collision
- Measurement error variance is σ^2 .
- $p_y(\sigma)$ = non-collision probability, given y

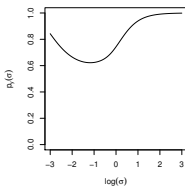
■ Note: large σ implies large $p_y(\sigma)$, *no matter what y is!*



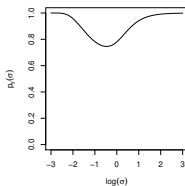
(a) $\|y\|^2 = 0.5$



(b) $\|y\|^2 = 0.9$



(c) $\|y\|^2 = 1.1$



(d) $\|y\|^2 = 1.5$

False confidence

- Even if satellite is on a direct collision course, if there's enough noise, then non-collision prob suggests it's safe
- *False confidence*: a hypothesis tending to be assigned large probability even though data does not support it

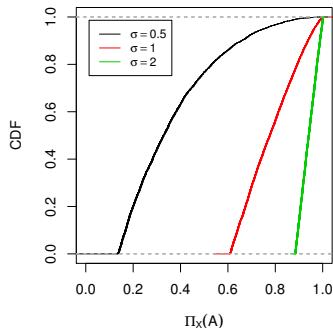
False confidence theorem (Balch, M., and Ferson 2019).

Let Q_Y be any continuous, Y -dependent probability dist'n on $\mathbb{T}$. For any $\alpha \in [0, 1]$, there exists $A \subset \mathbb{T}$ such that

$$A \not\subset \Theta \quad \text{and} \quad P_{Y|\Theta}\{Q_Y(A) > 1 - \alpha\} > \alpha.$$

- In words, there always exists false hypotheses that tend to be assigned high Q_Y -probability
- *Therefore, an IM whose output is probability can't be valid!*

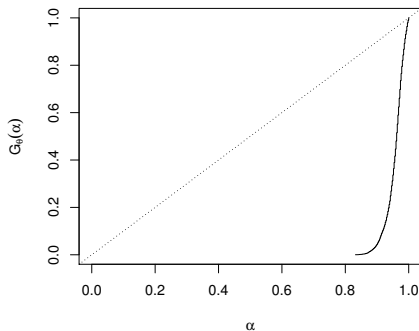
- Simplified version of the satellite example
 - let $A = \{\text{non-collision}\}$
 - then $Q_Y(A)$ as a random variable, with a CDF
 - plot CDF when data are generated under a collision course
- *False confidence*: $Q_Y(A)$ is almost always large!



- Example: Fieller–Creasy problem
- Model: $Y \sim P_{Y|\Theta} = N_2(\Theta, I)$, with $\Theta = (\Theta_1, \Theta_2)$
- Standard posterior: $Q_Y = N_2(y, I)$.
- Goal is to quantify uncertainty about the ratio $\Phi = \Theta_1/\Theta_2$
- Let $A = \{\theta : \theta_1/\theta_2 \leq 9\}$
- Suppose $\Theta = (1, 0.1)$, so that $\Phi = 10$ and A is false
- We'd expect $Q_Y(A)$ to be small
- Let's see...

False confidence, cont.

- Define the CDF: $G_\theta(\alpha) = P_{Y|\theta}\{Q_Y(A) \leq \alpha\}$.
- Plot uses $\theta = (1, 0.1)$, so $A = \{\theta_1/\theta_2 \leq 9\}$ is false
- Validity requires the CDF be on or above the diagonal line
- $Q_Y(A)$ actually tends to be *large* — false confidence!



- Let $\pi_Y(\theta)$ be a p-value for testing a point null about Θ .
- A point null about $\Phi = g(\Theta)$ is a composite null about Θ , so the former p-value is

$$\phi \mapsto \sup_{\theta: g(\theta)=\phi} \pi_Y(\theta).$$

- Similarly, a confidence region for Φ is

$$\left\{ \phi : \sup_{\theta: g(\theta)=\phi} \pi_Y(\theta) > \alpha \right\}.$$

- Notes:
 - the natural operation is *maximizing*
 - it's natural because it preserves the frequentist properties

- This leads to the following construction of a valid IM.
 - Define a function $\pi_y(\cdot)$ with the property that

$$Y \sim P_{Y|\Theta} \implies \pi_Y(\Theta) \sim \text{Unif}(0, 1).$$

- If $\sup_{\theta} \pi_y(\theta) = 1$ for all y , then set

$$\bar{\Pi}_Y(A) = \sup_{\theta \in A} \pi_Y(\theta).$$

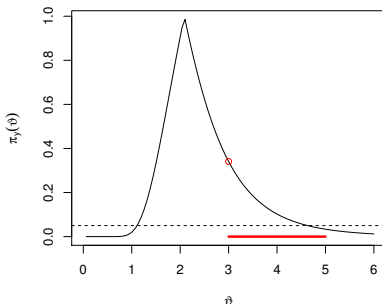
- This capacity has the mathematical form of a
 - *consonant plausibility function*⁷
 - or, equivalently, a *possibility measure*⁸
- Validity proof: $\Theta \in A$ and $\bar{\Pi}_Y(A) \leq \alpha \implies \pi_Y(\Theta) \leq \alpha$

⁷e.g., Shafer 1976, Ch. 10

⁸e.g., Dubois & Prade's work is extensive...

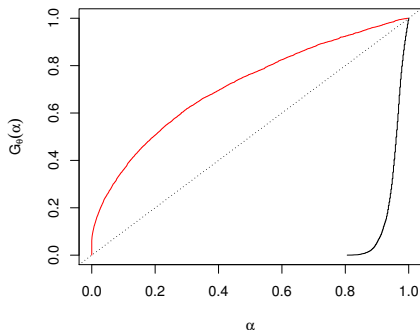
Frequentist to IM, cont. – visualization

- Contour π_y
- $A = [3, 5]$
- $\bar{\Pi}_y(A)$ via maximization
 - most plausible θ in A
 - size of A is irrelevant*
- $\{\theta : \pi_y(\theta) > 0.05\}$



* $\bar{\Pi}$ is a differential-element-free “measure” (M. [arXiv:2303.17425](https://arxiv.org/abs/2303.17425))

- Same Fieller–Creasy simulation as before
- Compare CDFs of $Q_Y(A)$ and $\underline{\Pi}_Y(A)$ ⁹
- Assertion A is false
- False confidence vs. **no false confidence**



⁹Based on contour $\pi_Y(\theta) = 1 - \text{pchisq}(\|y - \theta\|^2, 2)$

- Theorem implies validity can't be achieved with probability
- That is, the “most important unresolved problem” can't be resolved using the familiar tools
- Generalizations of probability theory are *needed*
- Fortunately, the imprecise probability framework that seem most suitable is the simplest one — *possibility theory*
- More formal characterizations can be given
 - roughly, for any procedure with nominal frequentist error rate control, there exists a valid IM (with the mathematical form described below) that's at least as efficient
 - see M. arXiv:2112.10904 for details

- Frequentist notions are difficult to interpret
- The imprecise prob connection gives a natural interpretation that's consistent with how they're used in practice
- That is, given the data, posited model, etc.
 - a p-value measures the *plausibility* of the null hypothesis
 - a confidence region is an aggregation of those sufficiently *plausible* parameter values¹⁰
- I teach this interpretation to my students:

"...your discussion of plausibility in ST503 made it easier for me to explain what a p-value is when I was interviewing. Thank you for the new perspective on the subject!"

¹⁰equivalently, a set with sufficiently large support

- Recap:
 - IMs are general constructs for probabilistic inference
 - defined the *validity* property
 - what mathematical structures are compatible with validity?
 - probability isn't, consonant plausibility is
- Shows how important imprecise prob is to statistical inference, needed to resolve “most important unresolved problem”
- **Part 2** focuses more on these details
 - what does imprecise prob look like?
 - how to construct a valid IM?
 - statistical examples?

- *Why should we care about false confidence?*
- Something won't necessarily go wrong, but there's a risk
- Statisticians directly involved in the data analysis might be able to manage their risk reasonably well
- But this level of involvement is rare — academic statisticians develop methods/software to be used by others
- To manage risk under this increased exposure, either
 - inform users which hypotheses are dangerous
 - or avoid false confidence altogether

- Want to learn more about imprecise probability?
- I taught a PhD-level special topics course on it in Fall 2022
- Lecture notes and videos are on the course website



ST 790 (001) Fall 2022 Advanced Special Topics

Imprecise-Probabilistic Foundations of Statistics & Data Science

[https://wordpress-courses2223.wolfware.ncsu.edu/
st-790-001-fall-2022/](https://wordpress-courses2223.wolfware.ncsu.edu/st-790-001-fall-2022/)

End of Part 1

- In **Part 1** I tried to convince you that valid probabilistic inference requires imprecise probability
- That argument was based on an attempt to convert p-values and confidence regions into valid inferential models (IMs)
- But that's not the only way; IMs can be constructed directly
- In **Part 2**, I'll describe
 - the general construction of a valid IM
 - and how to fine-tune it for efficiency

- basic IM construction
- validity theorem
- examples
- beyond validity: efficiency
- dimension reduction strategies:
 - conditioning
 - marginalization
- concluding remarks

- Observable data: Y in a sample space $\mathbb{Y}$
- Data are assumed to be relevant to the phenomenon under investigation, formalized through a *model*
- Statistical model: a collection of probability distributions

$$\mathcal{P} = \{P_{Y|\theta} : \theta \in \mathbb{T}\}$$

- Goal is to quantify uncertainty about the unknown Θ based on observed value $Y = y$ of the data.

Definition.

An *inferential model* (IM) is a mapping from $(y, \mathcal{P}, \dots)$ to a data-dependent capacity $\underline{\Pi}_y : 2^{\mathbb{T}} \rightarrow [0, 1]$ such that, for each $A \subseteq \mathbb{T}$, $\underline{\Pi}_y(A)$ represents ones degree of support in A , given data, etc.

- Quantify uncertainty via an inferential model (IM)
- We want that IM to be reliable or *valid*
- **Part 1:** *validity requires imprecision*
- Specifically: consonant beliefs/possibility measures
- Question: *how to construct such a thing directly?*

- Express the statement “ $Y \sim P_{Y|\Theta}$,” as

$$Y = a(\Theta, U), \quad U \sim P_U$$

- P_U (supported on $\mathbb{U}$) and a are known
- This describes how to simulate from the model, but also provides some relevant insight for inference
- Intuition:¹¹ *If U were observable*, then just solve for Θ in terms of the observed Y and U values — problem solved!
- Unfortunately, U is *not observable*...
- The distribution of U is known, so we can “guess” its unobserved value with some degree of reliability.

¹¹Chap. 4 in the book; also, [arXiv:1206.4091](https://arxiv.org/abs/1206.4091)

- The relation “ $Y = a(\Theta, U)$ ” is called an *association*
- Define the set-valued mapping

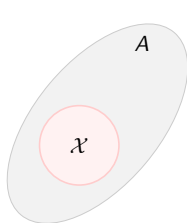
$$(y, u) \mapsto \mathbb{T}_y(u) = \{\theta \in \mathbb{T} : y = a(\theta, u)\}$$

- Intuition:
 - if $Y = y$ were observed and an oracle told me that $U = u$
 - then I would *know* “ $\Theta \in \mathbb{T}_y(u)$ ”
 - and inference is done, no induction needed
- Obviously, we can't rely on an oracle...

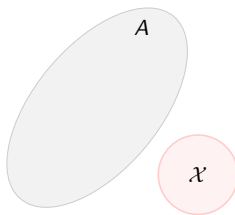
Construction: P-step

- Goal is to *guess* or *predict* the unobserved value of U
- Critical points:
 - uncertainty about a new draw from P_U would best be quantified by the *a priori* distribution P_U
 - unobserved value of U has a different status, it's fixed but unknown, so P_U isn't right *a posteriori*
 - this status change + our need for reliability implies that a different UQ framework is needed
- The P-step proceeds to guess/predict the unobserved value of U with a *random set*, $\mathcal{S} \sim P_{\mathcal{S}}$, taking values in $2^{\mathbb{U}}$

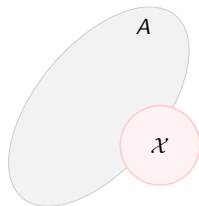
Brief interlude — random sets



(e) $\mathcal{X} \subseteq A$



(f) $\mathcal{X} \subseteq A^c$



(g) Otherwise

- Let $\mathcal{X}$ be a random set in $\mathbb{T}$
- Define the capacity¹² $\gamma(A) = \text{Prob}(\mathcal{X} \subseteq A)$
- Can you see why $\gamma(A) + \gamma(A^c) \leq 1$?

¹²This is a very special kind of capacity, ∞ -monotone, lots of structure

- *Combine* the output of the A- and P-steps, with the observed data $Y = y$, to get a new random set

$$\mathbb{T}_y(\mathcal{S}) = \bigcup_{u \in \mathcal{S}} \mathbb{T}_y(u)$$

- The *IM output* is the distribution of $\mathbb{T}_y(\mathcal{S})$ wrt $\mathcal{S} \sim P_{\mathcal{S}}$

- Key summaries:¹³

- Support/lower probability

$$\underline{\Pi}_y(A) = P_{\mathcal{S}}\{\mathbb{T}_y(\mathcal{S}) \subseteq A\}, \quad A \subseteq \mathbb{T}.$$

- Plausibility/upper probability

$$\overline{\Pi}_y(A) = 1 - \underline{\Pi}_y(A^c) = P_{\mathcal{S}}\{\mathbb{T}_y(\mathcal{S}) \cap A \neq \emptyset\}, \quad A \subseteq \mathbb{T}$$

- The plausibility contour is also important:

$$\pi_y(\theta) = P_{\mathcal{S}}\{\mathbb{T}_y(\mathcal{S}) \ni \theta\}, \quad \theta \in \mathbb{T}$$

- In certain cases, contour π_y determines the IM

¹³Here I'm assuming that $\mathbb{T}_y(\mathcal{S}) \neq \emptyset$ with $P_{\mathcal{S}}$ -probability 1; this almost always holds and, if it doesn't, then there's ways around it

- $(\underline{\Pi}_y, \overline{\Pi}_y)$ are dual capacities, $\overline{\Pi}_y(A) = 1 - \underline{\Pi}_y(A^c)$
- Special structure
 - belief & plausibility functions
 - often has even more structure, *consonance*
- $(\underline{\Pi}_y, \overline{\Pi}_y)$ determine a *credal set*¹⁴

$$\mathcal{C}(\overline{\Pi}_y) = \{Q_y \in \text{probs}(\mathbb{T}) : Q_y(\cdot) \leq \overline{\Pi}_y(\cdot)\}$$

- Reasoning with the IM output $(\underline{\Pi}_y, \overline{\Pi}_y)$ is equivalent to (probabilistic) reasoning with the collection $\mathcal{C}(\overline{\Pi}_y)$

¹⁴Non-empty, closed, convex set of probability dist'ns

- Similar to Fisher's fiducial argument from the 1930s and Dempster's extension from the 1960s
- The user-specified $\mathcal{S}$ is unique to our construction
- In particular, if $\mathcal{S}$ is a random singleton, i.e.,

$$\mathcal{S} = \{U\}, \quad U \sim P_U,$$

then the IM output agrees with that of Dempster's approach

- But validity is not guaranteed with singleton $\mathcal{S}$
- Effect of choosing non-singleton sets:
 - non-zero gap between lower and upper probabilities
 - gap implies non-additivity, critical to validity

Validity theorem

- Recall, validity ensures that, e.g., support assigned to false assertions tends not to be large
- Relevant to reliability of UQ

Definition.

An IM is *valid* if, for all $A \subseteq \mathbb{T}$ and all $\alpha \in [0, 1]$,

$$\sup_{\Theta \notin A} P_{Y|\Theta} \{ \underline{\Pi}_Y(A) > 1 - \alpha \} \leq \alpha$$

or, equivalently,

$$\sup_{\Theta \in A} P_{Y|\Theta} \{ \overline{\Pi}_Y(A) \leq \alpha \} \leq \alpha$$

Theorem.

With a “suitable choice of $\mathcal{S} \sim P_{\mathcal{S}}$,” the IM above is valid

- The IM can be used to design testing and confidence procedures just like discussed above
- Practical consequences of validity:
 - “Reject $H_0 : \theta \in A$ iff $\bar{\Pi}_y(A) \leq \alpha$ ” is a size α test
 - $\{\theta \in \mathbb{T} : \pi_y(\theta) > \alpha\}$ is a $100(1 - \alpha)\%$ confidence region

“Suitable choice of $\mathcal{S}$ ”

- For some intuition, what do we want $\mathcal{S}$ to do?
- Let $u_{y,\Theta}$ denote the unobserved value of U corresponding to the observed $Y = y$ and unknown Θ
- Recall: $\mathbb{T}_y(\mathcal{S}) = \bigcup_{u \in \mathcal{S}} \mathbb{T}_y(u)$
- Key insight: $\mathbb{T}_y(\mathcal{S}) \ni \Theta \iff \mathcal{S} \ni u_{y,\Theta}$
- *A priori*, however, $u_{y,\Theta}$ is just a realization of $U \sim P_U$
- Therefore:
 - if I choose $\mathcal{S}$ to be “good” at predicting realizations from P_U ,
 - then, e.g., $\pi_y(\Theta)$ is unlikely to be too small

Definition.

Given $\mathcal{S} \sim P_{\mathcal{S}}$, define $f(u) = P_{\mathcal{S}}(\mathcal{S} \ni u)$. Then $\mathcal{S}$ is *valid* if

$$U \sim P_U \implies f(U) \text{ is stochastically } \geq \text{Unif}(0, 1),$$

i.e., $P_U\{f(U) \leq x\} \leq x$ for all $x \in [0, 1]$

- The intuition from the previous slide(s) implies there must be some connection between $P_{\mathcal{S}}$ and P_U .
- Validity makes this connection mathematically precise
- Two extreme cases:
 - $\mathcal{S} \equiv \mathbb{U}$, valid
 - $\mathcal{S} = \{U\}$ with $U \sim P_U$, not valid
- The above condition is actually easy to arrange...

Lemma.

Let $h : \mathbb{U} \rightarrow \mathbb{R}$ be a function such that the distribution of $h(U)$ is non-atomic. Then the random set

$$\mathcal{S} = \{u \in \mathbb{U} : h(u) \leq h(U)\}, \quad U \sim P_U,$$

is valid in the sense above

- Common example is $P_U = \text{Unif}(0, 1)$:
 - good “default” choice
 - $h(u) = |u - 0.5|$
- Later we’ll see that there are reasons to take h to be bowl-shaped, leading to nested random sets
- This is not the only way to get a valid $\mathcal{S}$...

- Let $Y \sim P_{Y|\Theta} = \text{Bin}(n, \Theta)$, distribution function F_Θ .
- Construct a valid IM for Θ :
 - A. $F_\Theta(Y - 1) \leq U < F_\Theta(Y)$, $U \sim \text{Unif}(0, 1)$.
 - P. Simple (but not best) “default” random set

$$\mathcal{S} = \{u : |u - 0.5| \leq |U - 0.5|\}, \quad U \sim \text{Unif}(0, 1).$$

- C. Combine to get¹⁵

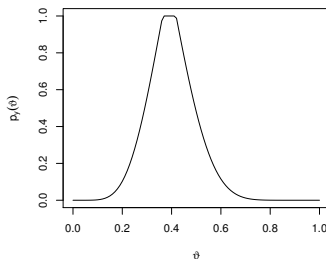
$$\begin{aligned}\mathbb{T}_y(\mathcal{S}) &= \bigcup_{u \in \mathcal{S}} \{\theta : F_\theta(y - 1) \leq u < F_\theta(y)\} \\ &= [1 - G_{n-y+1, y}^{-1}(\tfrac{1}{2} + |U - \tfrac{1}{2}|), 1 - G_{n-y, y+1}^{-1}(\tfrac{1}{2} - |U - \tfrac{1}{2}|)]\end{aligned}$$

- Distribution of $\mathbb{T}_y(\mathcal{S})$ only depends on $U \sim \text{Unif}(0, 1)$...

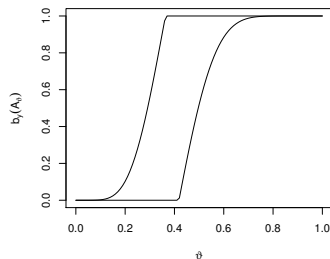
¹⁵Use fact that $F_\theta(y) = G_{n-y, y+1}(1 - \theta)$, beta distribution.

Examples: binomial, cont.

- Data: $n = 18$, $y = 7$.
- Plots of plausibility contour $\theta \mapsto \pi_y(\theta)$ and of the lower and upper probabilities of $A_\theta = [0, \theta]$.



(h) Plausibility contour



(i) lower/upper probabilities

Examples: general scalar problem

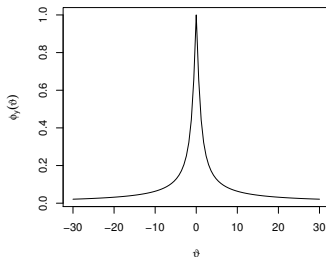
- Problems first considered by Fisher:
 - Scalar $Y \sim P_{Y|\Theta}$ and scalar Θ
 - continuous distribution function F_Θ
 - range of $F_\Theta(y)$ unconstrained by fixed y or fixed Θ .
- IM construction:
 - A. $Y = F_\Theta^{-1}(U)$, $U \sim P_U = \text{Unif}(0, 1)$
 - P. Default $\mathcal{S} = \{u \in [0, 1] : |u - 0.5| \leq |U - 0.5|\}$, $U \sim P_U$
 - C. $\mathbb{T}_y(\mathcal{S}) = \{\theta : F_\theta(y) \in \mathcal{S}\}$, with

$$\pi_y(\theta) = P_{\mathcal{S}}\{\mathbb{T}_y(\mathcal{S}) \ni \theta\} = 1 - |2F_\theta(y) - 1|$$

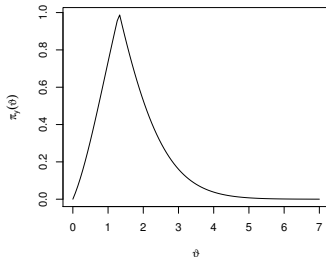
- Lots of examples can be covered by this analysis

Examples: general scalar problem, cont.

- Two cases:
 - $P_{Y|\Theta} = \text{Cauchy}(\Theta, 1)$
 - $P_{Y|\Theta} = \text{Gamma}(\Theta, 1)$.
- Plots below show the plausibility contour, $\pi_y(\theta)$.



(j) $y = 0$, Cauchy



(k) $y = 1$, gamma

Examples: multi-parameter

- n iid observations from $P_{Y|\Theta} = N(\mu, \sigma^2)$, $\Theta = (\mu, \sigma^2)$.
- Start with sufficient statistics $(\hat{\mu}, \hat{\sigma}^2)$; justification later.
- IM construction:
 - A. Let $(U_1, U_2) \stackrel{\text{iid}}{\sim} \text{Unif}(0, 1)$, $G = (n-1)^{-1}\text{ChiSq}(n-1)$, and $\Phi = N(0, 1)$. Then

$$\hat{\mu} = \mu + \sigma n^{-1/2} \Phi^{-1}(U_1) \quad \text{and} \quad \hat{\sigma}^2 = \sigma^2 G^{-1}(U_2).$$

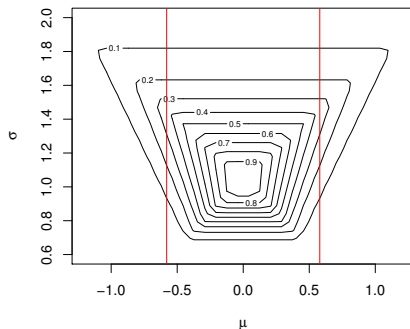
- P. Nested, 2-dim analogue of the “default”

$$\mathcal{S} = \left\{ (u_1, u_2) : \max_{i=1,2} |u_i - 0.5| \leq \max_{i=1,2} |U_i - 0.5| \right\}.$$

- C. Sort of messy...

Examples: multi-parameter, cont.

- Data: $n = 10$ and $(\hat{\mu}, \hat{\sigma}^2) = (0, 1)$.
- Plot shows:
 - joint plausibility contour for (μ, σ^2)
 - red lines: (marginal) 90% Student-t interval for μ
- This IM is valid but inefficient — to match the best inference on μ , we need to do formal marginalization...



- I've focused mainly on validity so far, but this isn't the only practical consideration
- Recall:
 - validity can be achieved simply by taking $\mathcal{S} \equiv \mathbb{U}$
 - results in $\overline{\Pi}_y(A) \equiv 1$ for all $A \neq \emptyset$
- Obviously, an IM with vacuous output is useless
- So it amounts to a balancing act:
 - validity requires $\mathcal{S}$ to be sufficiently large (in some sense)
 - but being too large causes inefficiency
- The goal is to take $\mathcal{S}$ as small as possible (in some sense) while maintaining validity

- Some progress on efficiency has been made
- One important result:¹⁶
 - A random set is *nested* if, for any pair of realizations, one is a subset of the other
 - *Theorem*. For any valid random set, there exists another that's valid, nested, and no less efficient
 - Therefore, I only use **nested random sets**
 - If $\mathcal{S}$ is nested, then IM output is *consonant*¹⁷
- I don't want to go through these details now
- There's a more important and related consideration concerning the *dimension* of the auxiliary variable

¹⁶Theorem 4.3 in the book

¹⁷Mathematically, that $\sup_{\theta \in \mathbb{T}} \pi_y(\theta) = 1$ for all y

- *Curse of dimensionality*: roughly, everything is more difficult in high dimensions than it is in low dimensions
- Suppose data $Y = (Y_1, \dots, Y_n)$ consists of n iid observations, with distribution depending on a low-dim Θ
- Naive association “ $Y = a(\Theta, U)$ ” involves a n -dimensional auxiliary variable $U = (U_1, \dots, U_n)$
- Often n will be large, which presents the challenge of guessing the high-dim U with a random set
- However:
 - it is generally possible to reduce the U dimension
 - simplifies the job of choosing valid/efficient random sets

- As a simple illustration, suppose we have n data points from a location model, i.e., $Y_i = \Theta + U_i$, $i = 1, \dots, n$
- After some algebraic manipulations, the above system of equations is equivalent to

$$\begin{aligned}\bar{Y} &= \Theta + \bar{U} \\ Y_i - \bar{Y} &= U_i - \bar{U}, \quad i = 1, \dots, n\end{aligned}$$

- Key observation: while U is unobservable, *there is a feature of U , namely $\{U_i - \bar{U} : i = 1, \dots, n\}$, that is observable*
- Consequences:
 - make a change of variable $U \mapsto \bar{U}$ to reduce dimension
 - use info about observed features to sharpen prediction of $\bar{U}$

- Generalize this idea...¹⁸
- Suppose there exists bijections $y \mapsto (T(y), H(y))$ and $u \mapsto (\tau(u), \eta(u))$ such that “ $Y = a(\Theta, U)$ ” is equiv to

$$T(Y) = b(\Theta, \tau(U)) \quad \text{and} \quad H(Y) = \eta(U)$$

- Want $T(y)$ and $\tau(u)$ to have same dimension as Θ
- Key point: the second expression has no Θ in it, hence the value of $\eta(U)$ is *observable*
 - We don't need to guess $\eta(U)$, hence dimension reduction
 - Observed $\eta(U)$ helps improve guess of unobserved $\tau(U)$
- Conditional association:

$$T = b(\theta, V), \quad V \sim P_{\tau(U)|\eta(U)=H(y)}$$

¹⁸Chap. 6 in book; also arXiv:1211.1530

- How to find (T, H) and (τ, η) ?
- Sufficient statistics:
 - For regular problems, min suff stat T has same dim as Θ
 - Just use an association for T , no conditioning needed.
- Group invariance:
 - T is an equivariant estimator and $\tau = T$
 - H is a maximal invariant and $\eta = H$.
- Solving a partial differential equation...

- Let $u_{y,\theta}$ be a solution to the equation $y = a(\theta, u)$.
- $\eta(U)$ would be observed if $\eta(u_{y,\theta})$ is constant in θ
- That is, we seek η that satisfies

$$\frac{\partial \eta(u_{y,\theta})}{\partial \theta} = 0$$

- We didn't know about existence in general,¹⁹ but sometimes we can solve it, e.g., method of characteristics²⁰
- Useful in non-regular problems when sufficiency falls short

¹⁹Recent progress in Zhang, Zhu, and Liu, arXiv:2301.05135

²⁰https://en.wikipedia.org/wiki/Method_of_characteristics

■ Uniform model²¹

- $X = (X_1, \dots, X_n) \stackrel{\text{iid}}{\sim} \text{Unif}\{a(\theta), a(\theta) + b(\theta)\}$
 - functions a and b known, θ unknown
 - suff stat: $Y = (Y_1, Y_2)$, $Y_1 = \min X_i$ and $Y_2 = \max X_i$
 - non-regular since Y is bivariate but θ is scalar
- Special case: $a(\theta) = \theta$ and $b(\theta) \equiv 1$
 - Association: $Y_i = \theta + U_i$, $U = (U_1, U_2)$ uniform order stat's
 - Since $u_{y,\theta} = y - \theta$, the PDE to solve is

$$0 \stackrel{\text{set}}{=} \frac{\partial \eta(u)}{\partial u} \Big|_{u=y-\theta} \cdot \frac{\partial (y - \theta)}{\partial \theta} = - \frac{\partial \eta(u)}{\partial u} \Big|_{u=y-\theta} \cdot 1_2$$

- Any function of $u_2 - u_1$ solves this, e.g., $\eta(u) = u_2 - u_1$

²¹M. and Lin, arXiv:1608.06791

- Tweaked association:

$$\underbrace{Y_1}_{T(Y)} = \Theta + \underbrace{U_1}_{\tau(U)} \quad \text{and} \quad \underbrace{Y_2 - Y_1}_{H(Y)} = \underbrace{U_2 - U_1}_{\eta(U)}$$

- Conditional IM construction:

A. Reduced association:

- $T(Y) = \Theta + V$

- $V = \tau(U) \sim P_{\tau(U)|\eta(U)=H(y)} = \text{Unif}\{0, 1 - H(y)\}$

P. Write $V = \{1 - H(y)\}Z$ where $Z \sim \text{Unif}(0, 1)$; default $\mathcal{S}$

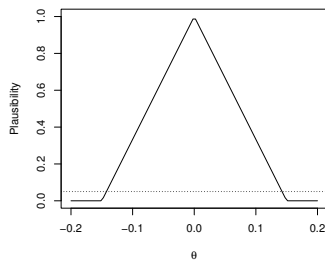
C. Plausibility contour

$$\pi_y(\theta) = 1 - \left| \frac{2}{1-H(y)} \{T(y) - \theta\} - 1 \right|$$

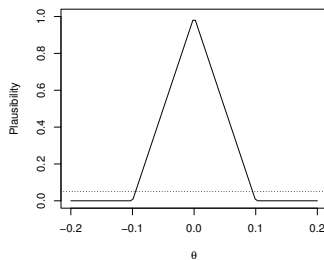
- e.g., IM's intervals for Θ agree with the Bayesian intervals,²² both are exact and efficient

²²Not a coincidence, see M. arXiv:2303.08630

- Uniform model
- Plots of plausibility contours for two data sets, $n = 10$



(l) $Y = (0.15, 0.85)$



(m) $Y = (0.10, 0.90)$

- *Curved normal model*

- $Y = (Y_1, \dots, Y_n)$ iid $N(\Theta, \Theta^2)$
- without (much) loss of generality, assume $\text{sign}(\Theta)$ is known
- classical non-regular problem, $\dim(\Theta) < \dim(\text{suff stat})$

- Obvious association with sufficient statistics:

$$\bar{Y} = \Theta + |\Theta| U_1 \quad \text{and} \quad S = |\Theta| U_2$$

- Two auxiliary variables for one parameter, inefficient!
- Rewrite association as

$$\bar{Y} = \Theta + S U_1 / U_2 \quad \text{and} \quad \underbrace{\frac{\bar{Y}}{S} = \frac{\text{sign}(\Theta) + U_1}{U_2}}_{\text{function of } U \text{ that's observed}}$$

- A. $\bar{Y} = \Theta + S V$ where V has the conditional distribution of U_1/U_2 , given **observed value of the function of U**
- P. Let f be the (conditional) density of V , and take $\mathcal{S}$ to be the random level sets

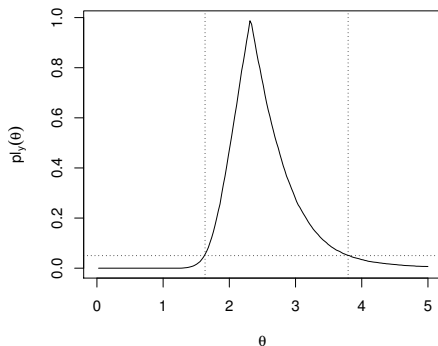
$$\mathcal{S} = \{v : f(v) \geq f(V)\}, \quad V \sim f$$

- C. For example, the plausibility contour is

$$\pi_Y(\theta) = P_{V|\dots\dots}\{f(V) \leq f(\frac{\bar{y}-\theta}{s})\}$$

Conditioning, cont.

- Experiment: $n = 10$ and $\Theta = 2$
- Data: $\bar{Y} = 1.97$ and $S = 2.48$
- Plausibility function $\pi_Y(\theta)$ and 95% plausibility interval



- IM is valid by construction, but what about efficient?
- Compare to a (generalized) fiducial solution which is (at least) first-order accurate asymptotically
- Repeat experiment above; table shows estimated coverage probability and mean lengths of 95% intervals

	Coverage probability	Mean length
Fiducial	0.932	1.45
IM	0.946	1.56

- All the “standard” examples can be covered by one or more of the techniques above
- What about “non-standard” examples, is there anything new that this perspective offers?
- Consider a so-called *non-regular* problem, where the dimension reduction via sufficiency is unsatisfactory
- An extension of the conditioning approach above can reduce dimension further, for valid and efficient inference
- Based on a *localization* technique

- Fix a localization point $t \in \mathbb{T}$
- Re-express the baseline association as

$$T(Y) = b(\theta, \tau(U)) \quad \text{and} \quad H_t(Y) = \eta_t(U).$$

- Construct a t -dependent IM which is valid at $t = \Theta$
- Specifically:
 - let $\pi_y(\cdot | t)$ denote the t -dependent plausibility contour
 - then $\sup_{\Theta} P_{Y|\Theta}\{\pi_Y(\Theta | \Theta) \leq \alpha\} \leq \alpha$
- Assuming that $\sup_{\theta} \pi_y(\theta | \theta) = 1$ for each y , we can use consonance to construct a *fused* plausibility function

$$\overline{\Pi}_y^{(F)}(A) = \sup_{\theta \in A} \pi_y(\theta | \theta)$$

- *Theorem.* The fused IM is valid

- $X_i \stackrel{\text{iid}}{\sim} \text{Unif}\{a(\Theta), a(\Theta) + b(\Theta)\}$
- $Y = (Y_1, Y_2)$, sample min and max
- Set $Y_i = a(\Theta) + b(\Theta) U_i$, with $U = (U_1, U_2)$ order stat's
- $u_{y,\theta} = b(\theta)^{-1}\{y - a(\theta)\}$
- For linear a, b , no localization needed; otherwise, yes
- Classical example:²³ $\text{Unif}(\Theta, \Theta^2)$, with $\Theta > 1$
- Can solve the PDE and find the relevant conditional distributions,²⁴ but it's not worth fleshing out details here
- Take-away: the resulting IM is valid

²³Hannig et al (2015, *JASA*)

²⁴M. and Lin, [arXiv:1608.06791](#)

- Strong Bayes and fiducial solutions are available too, asymptotic 2nd-order accurate
- Mean lengths of 95% intervals show that IM is efficient too

Method	n	θ					
		2	5	10	25	50	100
Bayes	2	0.6260	3.170	7.610	21.00	43.40	88.70
	5	0.2690	1.190	2.770	7.54	15.50	31.60
	10	0.1360	0.588	1.350	3.67	7.52	15.30
	25	0.0552	0.234	0.535	1.44	2.96	6.03
Fiducial	2	0.6170	3.080	7.420	20.70	43.00	88.30
	5	0.2680	1.180	2.760	7.51	15.40	31.60
	10	0.1360	0.586	1.350	3.66	7.51	15.30
	25	0.0552	0.234	0.535	1.44	2.96	6.02
IM	2	0.6130	3.070	7.430	20.80	43.10	88.00
	5	0.2670	1.180	2.770	7.53	15.40	31.40
	10	0.1360	0.587	1.350	3.66	7.51	15.20
	25	0.0552	0.234	0.535	1.44	2.96	5.98

Table 2: Estimated mean lengths of the 95% Bayes, fiducial, and IM intervals in the $\text{Unif}(\theta, \theta^2)$ simulation study. Estimates are based on 8000 Monte Carlo samples.

- Classical example: bivariate standard normal, correlation Θ
- Non-regular: parameter is 1-dim but min suff stat is 2-dim
- Problem setup:
 - (X_{1i}, X_{2i}) are iid bivariate normal
 - set $Y_1 = \sum_{i=1}^n (X_{1i} + X_{2i})^2$ and $Y_2 = \sum_{i=1}^n (X_{1i} - X_{2i})^2$
 - then $Y_1 = (1 + \Theta)U_1$ and $Y_2 = (1 - \Theta)U_2$
- Rewrite the basic association as

$$\frac{Y_1}{Y_2} = \frac{1 + \theta}{1 - \theta} \frac{U_1}{U_2} \quad \text{and} \quad \underbrace{\frac{Y_1}{U_1} + \frac{Y_2}{U_2}}_{\text{doesn't break up right}} = 2$$

- No function of U that's observed...

- Fix $t \in [-1, 1]$ and define

$$\eta_t(u) = (1 + t) \log u_1 + (1 - t) \log u_2$$

$$H_t(y) = (1 + t) \log\{y_1/(1 + t)\} + (1 - t) \log\{y_2/(1 - t)\}$$

- Reduced association

$$\underbrace{\log(Y_1/Y_2)}_{T(Y)} = \underbrace{\log\{(1 + \Theta)/(1 - \Theta)\}}_{z(\Theta)} + \underbrace{\log(U_1/U_2)}_{V=T(U)}$$

- Let $F_{\theta,y}$ denote the conditional CDF of $V = T(U)$ given $\eta_\theta(U) = H_\theta(y)$ — depends on θ and y
- Fused IM's plausibility contour

$$\pi_y(\theta \mid \theta) = 1 - |1 - 2F_{\theta,y}\{T(y) - z(\theta)\}|$$

- Simulation study: randomly sampled Θ from $\{0, 0.3, 0.6, 0.9\}$
- Table below compares coverage probability and lengths of three 90% confidence intervals for Θ

n	Coverage probability			Expected length		
	IM	r^*	Bayes	IM	r^*	Bayes
10	0.896	0.845	0.880	0.66	0.61	0.62
25	0.895	0.867	0.883	0.42	0.40	0.41
50	0.907	0.897	0.907	0.30	0.30	0.30
100	0.903	0.888	0.896	0.21	0.21	0.21

- Dimension reduction via conditioning is an effective strategy for constructing a valid and efficient IM.
- Works for standard examples, and provides new insights on how to handle non-standard examples.
- More examples in:
 - Cheng, Gao, and M. (2014), [arXiv:1406.3521](#)
 - Chapter 8 in the book
 - M. and Lin (2016), [arXiv:1608.06791](#)
 - M. and Syring, *ISIPTA'19 Proceedings*

Dimension reduction: marginalization

- Consider a decomposition $\Theta = (\Phi, \Lambda)$ where
 - Φ is the *interest parameter*
 - Λ is the *nuisance parameter*
- This is a common situation, e.g., we often care about a mean more so than we do the variance
- If we only care about inference on Φ :
 - creates an opportunity for dimension reduction
 - improved efficiency
- This is called *marginalization*, and the goal is to construct a marginal IM for Φ

- Start with an association $Y = a(\Theta, U)$ for Θ , where U has the same dimension as θ , maybe post-conditioning
- Suppose it can be rewritten as

$$R(Y, \Phi) = c(\Phi, V_1) \quad \text{and} \quad d(Y, V_2, \Phi, \Lambda) = 0$$

- Suppose $d(\cdot)$ satisfies the following property:

$$\forall (y, v_2, \phi) \quad \exists \lambda \quad \text{such that } d(y, v_2, \phi, \lambda) = 0$$

- Marginal association for ϕ : $R(Y, \Phi) = c(\Phi, V_1)$
- Dimension reduction by *ignoring the d component*²⁵

²⁵Chap. 7 in the book; also, [arXiv:1306.3092](https://arxiv.org/abs/1306.3092)

- Let $Y = (Y_1, \dots, Y_n)$ be iid $N(\mu, \sigma^2)$, target $\phi = \mu$
 - Conditional association – $(\hat{\mu}, \hat{\sigma}^2)$ suff stat's

$$\hat{\mu} = \mu + \sigma n^{-1/2} U_1 \quad \text{and} \quad \hat{\sigma} = \sigma U_2,$$

where $U_1 \sim N(0, 1)$, $U_2 \sim \{(n-1)^{-1} \text{ChiSq}(n-1)\}^{1/2}$.

- Rewrite this as

$$n^{1/2}(\hat{\mu} - \mu)/\hat{\sigma} = U_1/U_2 \quad \text{and} \quad \hat{\sigma} = \sigma U_2.$$

- Ignore the second equation to get the marginal assoc

$$\hat{\mu} = \mu + \hat{\sigma} n^{-1/2} V, \quad V \sim t_{n-1}.$$

- Similar story for classical linear regression (Ch. 8 in book)
- P + C steps lead to valid and efficient UQ

- Fieller–Creasy: $Y_i \stackrel{\text{ind}}{\sim} N(\Theta_i, 1)$, $i = 1, 2$.
- Target is $\Phi = \Theta_1/\Theta_2$; $\Lambda = \Theta_2$ is nuisance.
- Following Chapter 7.3.2 in the book:
 - Baseline association

$$Y_1 = \Phi\Lambda + U_1 \quad \text{and} \quad Y_2 = \Lambda + U_2.$$

- Rewrite it as

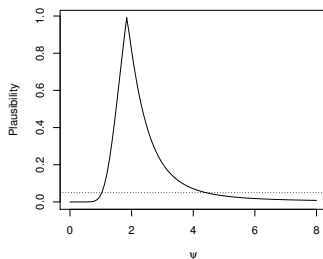
$$\underbrace{\frac{Y_1 - \Phi Y_2}{(1 + \phi^2)^{1/2}}}_{R(Y, \Phi)} = \underbrace{\frac{U_1 - \Phi U_2}{(1 + \Phi^2)^{1/2}}}_{= V \sim N(0,1)} \quad \text{and} \quad \underbrace{Y_2 - \Lambda - U_2}_{d(\cdots) = 0, \text{ ignore!}} = 0.$$

- Simple reduced association

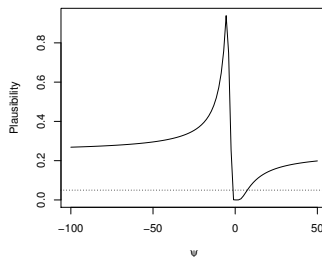
$$\frac{Y_1 - \Phi Y_2}{(1 + \Phi^2)^{1/2}} = V, \quad V \sim P_V = N(0, 1).$$

Marginalization, cont.

- Plots show plausibility contours for Φ in two data sets
- Second plot has a very unusual shape:
 - plausibility interval is unbounded
 - this feature is necessary for validity²⁶
- This IM solution matches the best possible.



(n) $y = (6.38, 3.46)$



(o) $y = (5.98, -1.16)$

²⁶Gleser & Hwang (1987), *Ann. Statist.*

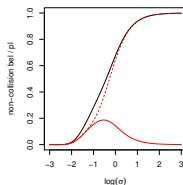
- Many-normal-means: $Y = \Theta + U$, where $U \sim N_n(0, I)$
- $\Theta = \Phi\Lambda$, where $\Phi = \|\Theta\|$ and $\Lambda = \Phi^{-1}\Theta \in \{\text{unit } n\text{-sphere}\}$
- $n = 2$ case $\iff$ simple satellite collision problem
- From Chapter 7.3.3 in the book, the reduced association is

$$\|Y\|^2 = F_{n,\phi}^{-1}(V), \quad V \sim P_V = \text{Unif}(0, 1),$$

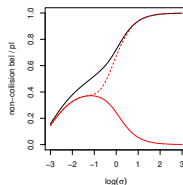
where $F_{n,\phi} = \text{ChiSq}(n; \phi^2)$

- Looks simple, but there are some constraints that need to be handled carefully — I won't discuss here

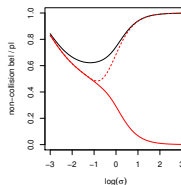
- Same satellite collision illustration as before, now with
 - $Q_y(\text{non-collision})$ (black)
 - $\underline{\Pi}_y(\text{non-collision})$ and $\overline{\Pi}_y(\text{non-collision})$ (dashed)
- Gap between support and plausibility is increasing in σ



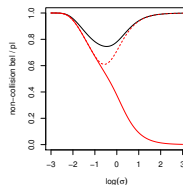
(p) $\|y\|^2 = 0.5$



(q) $\|y\|^2 = 0.9$



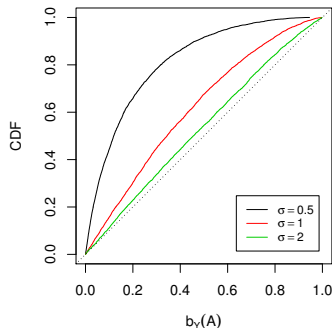
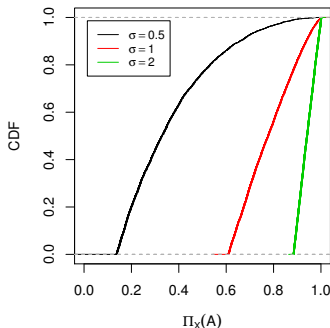
(r) $\|y\|^2 = 1.1$



(s) $\|y\|^2 = 1.5$

Marginalization, cont.

- Check validity by evaluating CDF of (lower) prob
- Same experiment as before
 - Left: $Q_Y(\text{non-collision})$ tends to be large
 - Right: $\Pi_Y(\text{non-collision})$ tends to be not large



- To recap:
 - marginal inference, when only Φ is of interest
 - dimension reduction improves efficiency
 - lots of non-trivial examples where this marginalization applies
- Sometimes the aforementioned condition on the d -component doesn't hold, e.g., Behrens–Fisher
- We developed a generalized marginal IM to handle B–F and other examples, but I won't discuss it here

- Conditioning can be difficult to implement
- Alternative strategy is to forget about writing an association that fully captures the sampling model
- Just write the association in terms of some low-dim summary
- In Ch. 9 of the book, we call this a *generalized IM*
- References:
 - M. (2015), arXiv:1203.6665
 - M. (2018), arXiv:1511.06733
 - Cahoon and M. (2020), arXiv:1910.00533
 - Cahoon and M. (2021), arXiv:1912.00037
 - Cella and M. (2022), arXiv:2112.10232

- Prediction is an extreme marginalization problem, i.e., all the model parameters are nuisance
- Can apply the above dimension reduction strategies to construct a valid IM for prediction
- New approach draws on the generalized IM idea and a connection to Shafer & Vovk's *conformal prediction*
- References:
 - M. and Lingham (2016), arXiv:1403.7589²⁷
 - Cella and M. (2021), arXiv:2001.09225
 - Cella and M. (2022), arXiv:2112.10234

²⁷Also, Chap. 9 in the book

- Three-step construction of a valid IM:
 - A. *associate* the data and unknown parameters with an unobservable auxiliary variable
 - P. *predict* that unobserved value of the auxiliary variable with a valid random set
 - C. *combine* the results of the two previous steps to get an IM (data-dependent lower/upper probability) that is valid
- For efficiency, it may be necessary to do some dimension reductions before completing the A-step
- Proved successful in a number of non-trivial examples, but there are challenges:
 - *conceptual*
 - *computational*

- A few other remarks...
- Details of the aforementioned frequentist characterization theorem require the *fused IM* notions²⁸
- Direct connections between IMs and possibility theory are available,²⁹ more on this in **Part 3**
- What happens if $\Theta_y(\mathcal{S})$ can be empty?
 - one option is to use Dempster's rule of combination to eliminate conflict cases
 - another option is to manually “stretch” the random set³⁰
 - latter approach was shown to be more efficient, at least in certain cases involving non-trivial parameter constraints

²⁸M., arXiv:2112.10904

²⁹Liu and M., arXiv:2008.06874

³⁰Ermini Leaf and Liu (2012), *IJAR*

IMs for modern ML-type problems



ELSEVIER

International Journal of Approximate Reasoning

Volume 150, November 2022, Pages 1-18



Valid inferential models for prediction in supervised learning problems ☆

[Leonardo Cella](#)  , [Ryan Martin](#) 



ELSEVIER

International Journal of Approximate Reasoning

Volume 151, December 2022, Pages 205-224



Direct and approximately valid probabilistic inference on a class of statistical functionals

[Leonardo Cella](#) ^a  , [Ryan Martin](#) ^b 

End of Part 2

- I hinted at some “challenges” in the **Part 2** conclusion
- More specifically:
 - requires an association, shouldn't be necessary
 - requires specification of a random set, shouldn't be necessary
 - people struggle to understand random sets, etc
- *Question*: Is there a more direct construction?
- Yes, but requires some new perspectives, and that's what I want to share with you here
- These efforts are worthwhile — big picture implications...

Part 3: Outline

- big picture
- key insight — spectrum of problems
- new IM construction
- properties
- practicalities & examples
- brief thoughts on regularization
- concluding remarks

- *Bayesian, frequentist, fiducial, ...*
- A variety of theories is great for “publish-or-perish” academia, but not for science/society/etc
- Fundamental questions remain unanswered:
Should probability enter to capture degrees of belief about claims? ... Or to ensure we won't reach mistaken interpretations of data too often in the long run...? —Mayo
- ASA³¹ offers non-answers, e.g., “Different measures of uncertainty can complement one another”
- Lack of answers → confusion, distrust, and embarrassment

³¹<https://magazine.amstat.org/blog/2021/08/01/task-force-statement-p-value/>

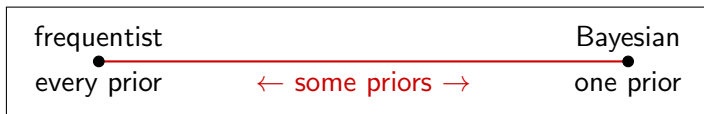
How can a discipline, central to science and to critical thinking, have two methodologies, two logics, two approaches that frequently give substantially different answers to the same problems. Any astute person from outside would say, “Why don’t they put their house in order?” ... [T]he two approaches have been around since 1763 and 1930 with regular disagreement and yet no sense of urgency to clarify the conflicts. And now even a tired discipline can just ask, “Who wants to face those old questions?” ... But is complacency in the face of contradiction acceptable for a central discipline of science? —Fraser

- Foundational questions, e.g., Bayesian vs frequentist, aren't just impractical details to philosophize about
- But you don't have to take my word for it...

Two contending philosophical parties, the Bayesians and the frequentists, have been vying for supremacy over the past two-and-a-half centuries... Unlike most philosophical arguments, this one has important practical consequences. The two philosophies represent competing visions of how science progresses and how mathematical thinking assists in that progress. —Efron

Key insight

- We tend to think in binary terms:
 - prior $\iff$ Bayesian
 - no prior $\iff$ frequentist
- However, frequentist isn't "no prior" — it's *every prior*
- So, there's a spectrum indexed by **partial prior** info



- Let $\mathcal{Q}$ denote a (closed & convex) set of priors for Θ
 - $\mathcal{Q} = \{\text{all the priors}\}$, frequentist
 - $\mathcal{Q} = \{\text{one single prior}\}$, Bayesian
 - *all the other (relevant) cases no one's considered*

- Define an “upper joint distribution” for (Y, Θ)

$$\bar{P}_{\mathcal{Q}}(Y \in B, \Theta \in A) = \sup_{Q \in \mathcal{Q}} \underbrace{\int_A P_{Y|\theta}(Y \in B) Q(d\theta)}_{\text{linear functional of } Q}$$

- “Frequentist corresponds to every prior”?
 - e.g., set estimator $C(Y)$ with $\sup_{\Theta} P_{Y|\Theta}\{C(Y) \not\equiv \Theta\}$ small
 - “Boundary” of $\mathcal{Q} = \{\text{all priors}\}$ consists of point masses
 - Maximum of a linear functional over a closed & convex set is attained on the boundary, i.e.,

$$\begin{aligned}\bar{P}_{\mathcal{Q}}\{C(Y) \not\equiv \Theta\} &= \sup_{\text{all } Q} \int P_{Y|\theta}\{C(Y) \not\equiv \theta\} Q(d\theta) \\ &= \sup_{\Theta} P_{Y|\Theta}\{C(Y) \not\equiv \Theta\}\end{aligned}$$

- New n -part series of papers:

*Valid and efficient imprecise-probabilistic
inference with partial priors*

- Two parts of the series are completed so far

1 arXiv:2203.06703

what does *validity* look like in this case, what are its consequences, and how can it be achieved

2 arXiv:2211.14567

answers to the above questions in first part suggest a powerful new IM construction

- Working towards a new book...

- There are some key principles justifying what follows, but we're short on time here; roughly:
 - $\bar{P}_{\mathcal{Q}}$ determines an imprecise prob on $\mathbb{Y} \times \mathbb{T}$
 - can be bounded by a suitable possibility measure
 - such that y -slices are possibility measures on $\mathbb{T}$
- Workhorse: *imprecise-probability-to-possibility transform*³²
- Above principles suggest an IM with plausibility contour

$$\pi_y(\theta) = \bar{P}_{\mathcal{Q}}\{R_q(Y, \Theta) \leq R_q(y, \theta)\},$$

where

- $q(\theta) = \sup_{Q \in \mathcal{Q}} Q(\{\theta\})$, determined by $\mathcal{Q}$
- $R_q(y, \theta) = L_y(\theta) q(\theta) / \sup_{\vartheta} \{L_y(\vartheta) q(\vartheta)\}$

³²Hose 2022 PhD thesis; M. arXiv:2211.14567

- Contour above leads to a full possibility measure via

$$\overline{\Pi}_Y(A) = \sup_{\theta \in A} \pi_Y(\theta), \quad A \subseteq \mathbb{T}$$

- Dependence on $\mathcal{Q}$ is implicit
- Enjoys a (sort of) familiar property, with a number of important consequences

Validity theorem.

The new IM construction above is *strongly valid* (wrt $\mathcal{Q}$), i.e.,

$$\overline{P}_{\mathcal{Q}}\{\pi_Y(\Theta) \leq \alpha\} \leq \alpha, \quad \alpha \in [0, 1].$$

Corollary.

IM-based procedures are reliable in a $\mathcal{Q}$ -frequentist sense:

- 1** the test “reject $H_0 : \Theta \in A$ iff $\bar{\Pi}_Y(A) \leq \alpha$ ” satisfies

$$\bar{P}_{\mathcal{Q}}\{\underbrace{\bar{\Pi}_Y(A) \leq \alpha, \Theta \in A}_{\text{“Type I error”}}\} \leq \alpha, \quad \alpha \in [0, 1], \quad A \subseteq \mathbb{T}$$

- 2** the region $C_\alpha(y) = \{\theta : \pi_Y(\theta) > \alpha\}$ satisfies

$$\bar{P}_{\mathcal{Q}}\{C_\alpha(Y) \not\ni \Theta\} \leq \alpha, \quad \alpha \in [0, 1]$$

Corollary.

The error rate control in Item 1 above is *uniform in A*, i.e.,

$$\bar{P}_{\mathcal{Q}}\left[(Y, \Theta) : \bigcup_{A: A \ni \Theta} \{\bar{\Pi}_Y(A) \leq \alpha\}\right] \leq \alpha$$

- An imprecise probability like $\overline{\Pi}_y$ determines a (closed & convex) set of probabilities called its *credal set*

$$\begin{aligned}\mathcal{C}(\overline{\Pi}_y) &= \{Q_y \in \text{probs}(\mathbb{T}) : Q_y(\cdot) \leq \overline{\Pi}_y(\cdot)\} \\ &= \text{all probabilities on } \mathbb{T} \text{ dominated by } \overline{\Pi}_y\end{aligned}$$

- Credal sets of possibility measures have a very special characterization, namely,

$$Q_y \in \mathcal{C}(\overline{\Pi}_y) \iff \underbrace{Q_y\{\theta : \pi_y(\theta) > \alpha\}}_{\text{i.e., } Q_y \text{ is a confidence dist'n}} \geq 1 - \alpha$$

- In group invariant problems (e.g., location–scale), when the Fisher’s fiducial distribution is well-defined, it corresponds to the “maximal” element³³ in $\mathcal{C}(\overline{\Pi}_y)$

³³M. arXiv:2303.08630

- Subjective Bayesian view:
 - prior $Q(A)$ represents the original price at which they'd buy or sell gambles on " $\Theta \in A$ "
 - and $Q_y(A)$ is their updated price after observing $Y = y$
- Goal is for the update to be *coherent*, i.e., be such that the agent can't be made a sure-loser
- More complicated in the imprecise probability world³⁴
- Suffice it to say that the IM's "update" from $\mathcal{Q}$ to $\mathcal{C}(\bar{\Pi}_y)$ satisfies certain coherence properties³⁵
- For example,

$$\sup_{y \in \mathbb{Y}} \bar{\Pi}_y(A) \geq \sup_{Q \in \mathcal{Q}} Q(A), \quad \text{all } A \subseteq \mathbb{T}$$

³⁴see, e.g., Walley's theory of lower previsions

³⁵Section 5.2.2 in M. arXiv:2211.14567

- In general, $\pi_Y(\theta)$ is a Choquet integral
- When q is a possibility contour, which can often be justified, the Choquet integral formula simplifies to

$$\pi_Y(\theta) = \int_0^1 \left[\sup_{\vartheta: q(\vartheta) > s} P_{Y|\vartheta} \{ R_q(Y, \vartheta) \leq R_q(y, \theta) \} \right] ds$$

- Naive ideas:
 - approximate $P_{Y|\vartheta} \{ \dots \}$ via Monte Carlo — maybe using importance sampling — over a ϑ -grid³⁶
 - stochastic gradient descent³⁷
- How to do all this efficiently?

³⁶Hose, Hanss, and M. *BELIEF 2022 Proceedings*

³⁷Syring and M. [arXiv:2103.02659](https://arxiv.org/abs/2103.02659)

- Often there's simplifying structure in the problem
- *Principle of Minimal Complexity*:³⁸ Reduce dimension!
- Applications of PMC often involve *conditioning*
- Example:
 - frequentists: $\mathcal{Q} = \{\text{all priors}\}$
 - $R_q(y, \theta) = \text{relative likelihood}$
 - problem allows for *conditioning* on $\Theta = \theta$:

$$\pi_y(\theta) = P_{Y|\theta}\{R_q(Y, \theta) \leq R_q(y, \theta)\}$$

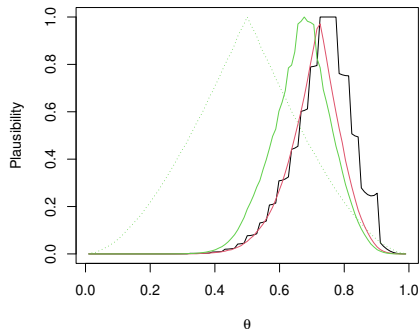
- this is familiar... it's a p-value for $H_0 : \Theta = \theta$
 - same validity properties, but more efficient
- Applies in other ways too...

³⁸M., arXiv:2211.14567

- Conceptually simple, not unfamiliar
- Strong theoretical support
- Computation is non-trivial, but tools are available
- No time to discuss marginalization in detail
 - sometimes can do conditioning
 - sometimes other tricks
 - profiling works in *most* other cases
- Basically any application is an open problem!

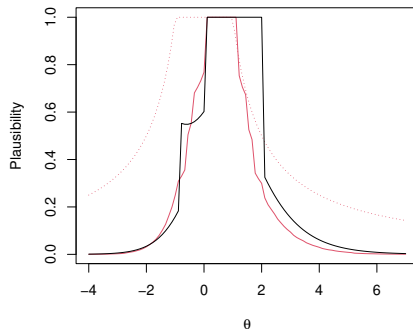
Examples

- Binomial model: $P_{Y|\theta} = \text{Bin}(n, \theta)$
- IM possibility contour ($n = 16, y = 12$)
 - frequentist “every prior”
 - Bayesian “one prior”
 - IM “partial prior”



Examples, cont.

- Two-sample binomial, Θ denotes the log odds ratio
- Data $(n_1, y_1) = (43, 1)$ and $(n_2, y_2) = (39, 2)$
- Marginal IM possibility contour³⁹
 - vacuous prior (black)
 - partial prior⁴⁰ (red)

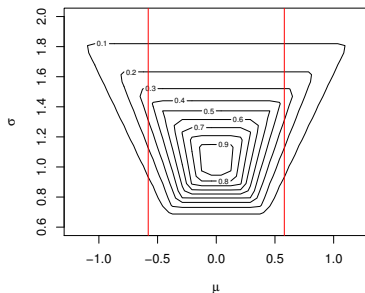


³⁹Nuisance parameters eliminated via conditioning

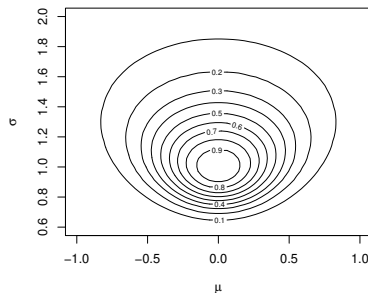
⁴⁰prior info " $E|\Theta| \leq 1$ " $\rightarrow q(\theta) = 1 \wedge |\theta|^{-1}$ (red dotted)

Examples, cont.

- Same normal illustration: $n = 10$, $(\hat{\mu}, \hat{\sigma}) = (0, 1)$
- Compare:
 - contour plot for the “old IM”
 - contour plot for the “new IM”



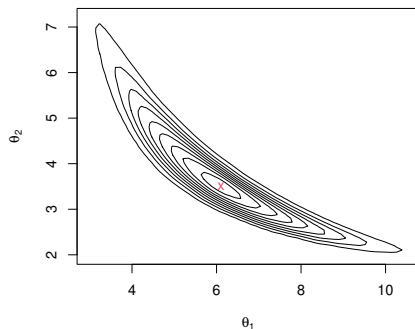
(t) Old version



(u) New version

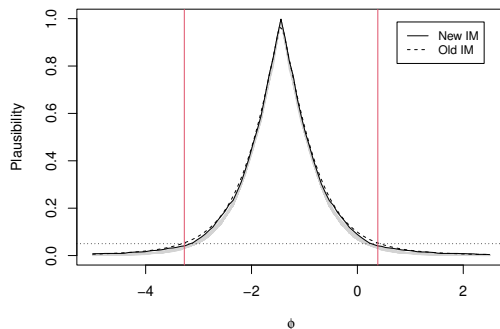
Examples, cont.

- Model $P_{Y|\theta} = \text{Gamma}^{\otimes n}(\theta_1, \theta_2)$
- Simulated data, $n = 25$, $\theta_1 = 7$ and $\theta_2 = 3$
- Map of the IM possibility contour (vacuous prior)
- Computation via importance sampling



Examples, cont.

- Behrens–Fisher problem (w/ vacuous prior)
- Data from Lehmann's famous travel time example
- Marginalization via profiling
- Numerically the same as marginal IM solution in the book⁴¹

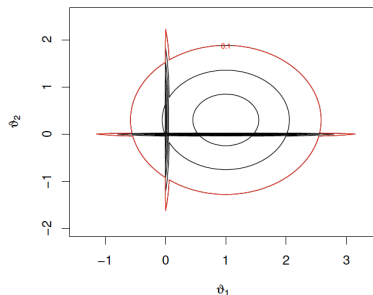
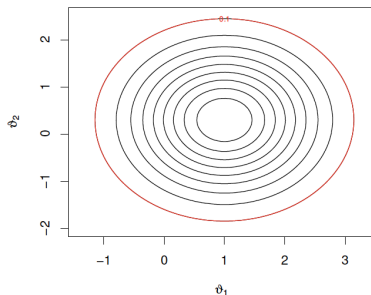


⁴¹I haven't been able to prove the equivalence...

- Regularization is incorporation of partial prior info
- That is, we're pretty sure the parameter vector is sparse, so put "something" in to enforce this (imprecise) constraint
- Can construct a partial prior contour that encodes sparsity or whatever low-dim structure is relevant
- All the above theory holds!
- Only questions are efficiency & computation
- Promising new direction...

Regularization, cont.

- $Y \sim N_2(\theta, I)$, observed $y = (1, 0.3)$
- Plausibility contours:
 - vacuous-prior (left)
 - sparsity encouraging partial-prior IM (right)
- 90% plausibility region — note efficiency gain on right



- ML applications don't have (parametric) models
- Above construction depends on model in
 - the ranking function " R_q "
 - the "prob" in my prob-to-poss transform
- But maybe we don't actually need those things...
- Let's focus on a prediction setting
 - $Y^n = (Y_1, \dots, Y_n)$ is observable, interest in Y_{n+1}
 - assume exchangeable with unknown dist'n P
 - set of probabilities, $\mathcal{P} = \{\text{exchangeable } P\}$
- Can handle covariates too, just messier

- Let $\gamma : \mathbb{Y}^n \times \mathbb{Y} \rightarrow \mathbb{R}$ be a measure of *conformity*
 - large values of $\gamma(y^n, y_{n+1})$ mean y_{n+1} “conforms with” y^n
 - e.g., $\gamma(y^n, y_{n+1}) = -d(\hat{y}_{z^n}, y_{n+1})$
 - assume invariant to permutations of its first argument
- Let's use γ in place of R_q
- Naive application of the prob-to-poss transform suggests

$$\pi_{y^n}(y_{n+1}) = \sup_{P \in \mathcal{P}} P\{\gamma(Y^n, Y_{n+1}) \leq \gamma(y^n, y_{n+1})\}$$

- Corresponding IM has all the nice properties, but...
- A practical solution requires a complexity reduction

- *Principle of Minimal Complexity*... conditioning
- Condition on the set of values⁴² $\{y_1, \dots, y_n, y_{n+1}\}$

$$\pi_{y^n}(y_{n+1}) = \sup_{P \in \mathcal{P}} P[\gamma(Y^n, Y_{n+1}) \leq \gamma(y^n, y_{n+1}) \mid \{y^n, y_{n+1}\}]$$

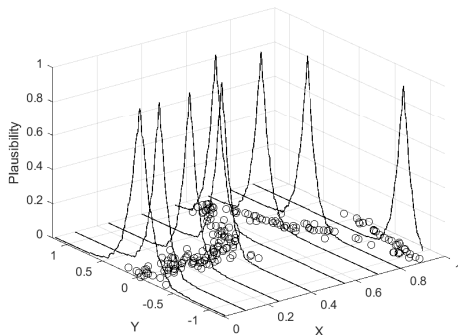
$$\text{sufficiency} \rightarrow = \frac{1}{(n+1)!} \sum_{\sigma} 1\{\gamma(y^{\sigma(1:n)}, y_{\sigma(n+1)}) \leq \gamma(y^n, y_{n+1})\}$$

$$\text{invariance} \rightarrow = \frac{1}{n+1} \sum_{i=1}^{n+1} 1\{\gamma(y_{-i}^{n+1}, y_i) \leq \gamma(y^n, y_{n+1})\}$$

- RHS is Vovk & Shafer's conformal prediction “transducer”

⁴²Faulkenberry (*JASA* 1973), Hoff (*Bernoulli* 2023)

- For feature-label pairs $Z = (X, Y)$, IM produces possibility contour functions for Y at each $X = x$
- Below is an illustration⁴³



⁴³Cella and M., arXiv:2112.10234

- Closing the Bayesian-vs-frequentist gap is important
 - frequentism ignores available info
 - Bayesianism embellishes on it⁴⁴
- Requires something new, and the proposed possibility-theoretic IM is a strong candidate solution
- More flexible than both Bayesian and frequentist because it can accommodate partial priors
- Most problems have partial prior info — notion is missing in the literature because classical theory can't handle it
- IM also has strong statistical and behavioral properties

⁴⁴Manski's *Law of Decreasing Credibility*: The credibility of inference decreases with the strength of the assumptions maintained

- Imprecision can help to achieve valid UQ, but that's not all:
 - model ambiguity and robustness
 - coarse or incomplete data
- Open questions:
 - efficient computational strategies (for high-dim cases)?
 - beyond parametric models?
 - model assessment?
 - partial prior elicitation/construction?
 - ...
- Relatively low-hanging-fruit applications:
 - mixture models
 - generalized linear models
 - censored data
 - ...
- I'm excited about the "possibilities" (pun intended)

End of Part 3



ST 790 (001) Fall 2022 Advanced Special Topics

Imprecise-Probabilistic Foundations of Statistics & Data Science

<https://wordpress-courses2223.wolfware.ncsu.edu/st-790-001-fall-2022/>

- Links to papers, talks, etc. can be found on my website:

www4.stat.ncsu.edu/~rmartin/

- If you have questions or just want to talk, please feel free to email me at rgmarti3@ncsu.edu

Thanks for your attention!